Deduce the World: How Far Can Reasoning Go?

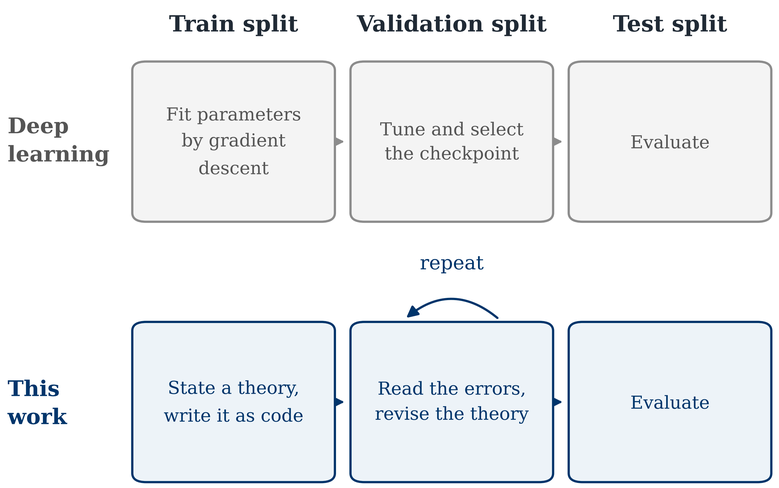
Jeffrey Wei — Yale University
Brian Cheung — University of California, San Francisco
Correspondence: jeffrey.wei@yale.edu

Results & figures

1 Replacing the learner

Science deduces the world from first principles, revising its theories where their predictions fail.

We ask whether a reasoning model can form scientific theories, and whether those theories generalize better than deep learning.

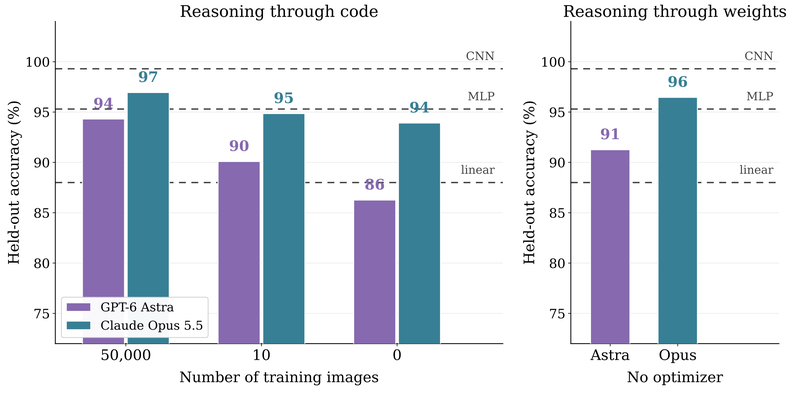


May read papers and the web, measure the training data, inspect its own errors

May not fit parameters to labels, search automatically over rules, store examples, see the test split

2 Classifying handwritten digits

Can a model classify handwritten digits from a stated theory of how digits are written, and how much data does it need to state one?

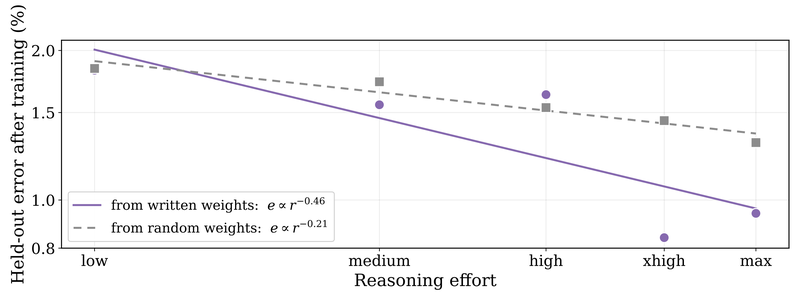


MNIST, 60,000 training and 10,000 test images. Best of five reasoning efforts; dashed lines are trained models.

Given no training images at all, a theory of how digits are written classifies most of them.

Reasoning as initialization

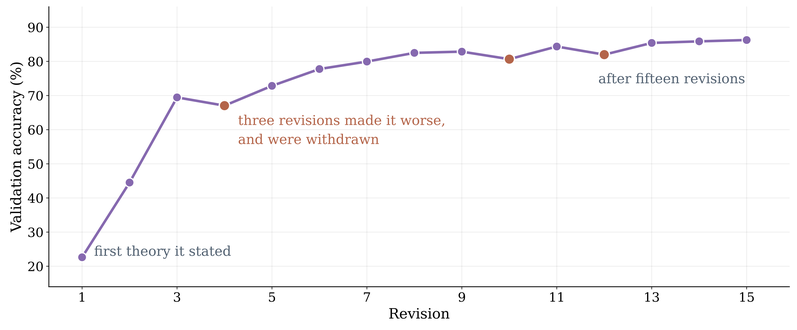
Can a model reason out an architecture and its weights directly, and is what it writes a better starting point than random initialization?



Networks initialized from written weights train to lower error than networks initialized at random, and the gap widens with reasoning effort.

The deduction–abduction loop

Can a model abduce why its own theory fails, and correct it?

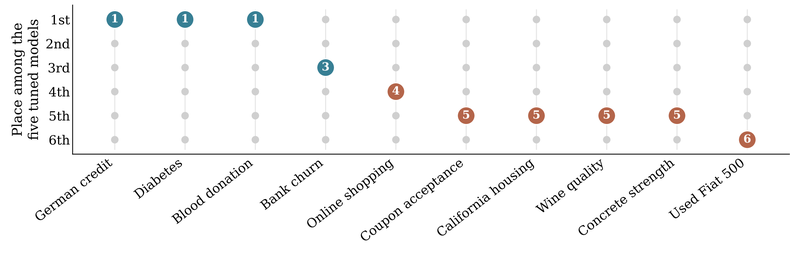


One digits session, every revision it made. Each began with a written hypothesis.

This is the scientific method: propose a cause, predict its effect, measure, and keep only what survives.

3 Reasoning on tabular data

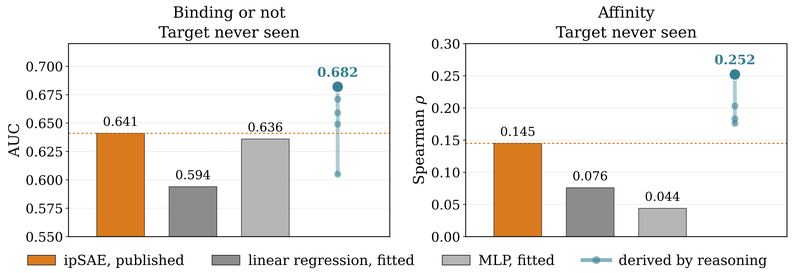
Does the same protocol compete on tabular data, where gradient-boosted trees are the default?



Ten tasks from TabArena. Each ranks the reasoned theory against a linear model, a random forest, XGBoost, CatBoost and an MLP, all tuned on the same rows.

4 Protein binding from first principles

Can a model derive a binding score from physics rather than fit one? Each session receives a structure predictor's error matrix and nothing else.



704 designed binders, holding out a whole target. Detection on the left, affinity on the right.

A rule derived from physics beats the published score on a target it has never seen, and beats both fitted models.

The theory it stated

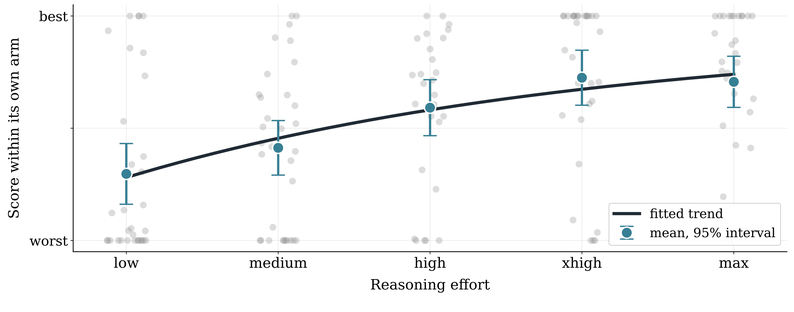
Treat the bound complex as a harmonic well. Equipartition makes the interface stiffness the inverse square of its fluctuation, and a stiffer well is a deeper one. Dividing by the binder's own internal error removes the predictor, and the units with it. Association confines six rigid-body degrees of freedom, each worth half the thermal energy times the log of the stiffness.

$$\kappa = \left(\frac{\sigma_0}{\sigma_{\text{iface}}}\right)^2, \qquad \mathrm{p}K_{\mathrm{d}} = \mathrm{p}K_{\mathrm{d,ref}} + 3\log_{10}\kappa.$$

The three is six halved. Nothing is fitted.

5 Scaling reasoning effort

Does more reasoning produce better theories?



6 So how far can reasoning go?

We hypothesize that as language model reasoning improves, we will develop stronger first principles in the theory-poor, data-rich domains where deep learning currently dominates.